

Evaluating Retrieval-Augmented Generation Pipelines: A Non-Invasive, Modular Benchmarking Framework for Automated RAG Configuration Analysis

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Abstract

Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for grounding large language model outputs in factual, domain-specific knowledge. Yet practitioners face a vast, poorly understood configuration space spanning chunking strategies, embedding models, retrieval methods, and generation models.

We address this gap with a modular benchmarking framework that attaches to existing RAG pipelines as a drop-in evaluation layer, wrapping their components without restructuring the pipeline. It systematically explores the configuration space across all components, replacing manual tuning.

Every benchmarked configuration is tracked end-to-end via MLflow for full reproducibility and evaluated through a hybrid scheme combining LLM-based quality metrics with established information-retrieval and natural-language-generation measures. Pre-configured dataset adapters cover diverse QA domains and are extensible to custom domain-specific data. We validate the framework by benchmarking more than 100 configuration combinations on locally served open-weight models, demonstrating the reproducibility and comparability of RAG components with minimal modifications to existing pipelines.